

Appendix B: LCA: main assumptions and selected background data from LCI databases

Table B.1: Overview of parameter definition for the circularity indicator

Life cycle aspect	Main assumption	Selected process from database
<i>Transport of materials to the manufacturing plant</i>	<i>This has been excluded because the contribution is assumed to be limited.</i>	-
Material production	The system boundary for the LCA is different compared to the PCI because all upstream processes related to material extraction and refining are included. Some modifications have been made to the selected processes: → amount of recycled content of feedstock have been modified to align with data presented in the paper → amount of precious metal modified to align with BOM presented in paper	Concrete block {GLO} market for Alloc Def, U Steel, unalloyed {GLO} market for Alloc Def, U Steel, low-alloyed, hot rolled {GLO} market for Alloc Def, U Steel, chromium steel 18/8, hot rolled {GLO} market for Alloc Def, U Cast iron {GLO} market for Alloc Def, U Aluminium, primary, ingot {RoW} market for Alloc Def, U Copper {GLO} market for Alloc Def, U Flat glass, uncoated {RER} production Alloc Def, U Polypropylene, granulate {GLO} market for Alloc Def, U Glass fibre reinforced plastic, polyamide, injection moulded {GLO} market for Alloc Def, U Acrylonitrile-butadiene-styrene copolymer {GLO} market for Alloc Def, U Synthetic rubber {GLO} market for Alloc Def, U Nylon 6-6 {GLO} market for Alloc Def, S Polyethylene, LDPE, granulate, at plant/RER Polymethyl methacrylate (PMMA) beads, production mix, at plant RER Printed wiring board, surface mounted, unspecified, Pb free {GLO} production Alloc Def, U

Life cycle aspect	Main assumption	Selected process from database
Product Manufacturing	Some modifications have been made to the selected processes: → amount of material losses during material processing have been modified to align with data presented in the paper	Metal working, average for metal product manufacturing {RoW} processing Alloc Def, U Injection moulding {GLO} market for Alloc Def, U Tempering, flat glass {GLO} market for Alloc Def, U
<i>Assembly phase</i>	<i>This has been excluded because the contribution is assumed to be limited and the same in all different scenario's</i>	-
<i>Transport and distribution of the device to the final user</i>	<i>This has been excluded because the contribution is assumed to be limited.</i>	-
Use phase — energy consumption	Based on available literature, for the use phase a energy consumption in real-life conditions of 0.672 kWh/wash cycle is assumed	Electricity, low voltage {GLO} market group for Alloc Def, U
<i>Use phase — water consumption</i>	<i>This has been excluded because the contribution is assumed to be limited.</i>	-
<i>Use phase — detergents</i>	<i>This has been excluded because the contribution is assumed to be limited and the same in all different scenario's</i>	-
<i>Transport of the device to the end-of-life facility</i>	<i>This has been excluded because the contribution is assumed to be limited.</i>	-

Life cycle aspect	Main assumption	Selected process from database
End of life processing	The uncollected products at end of life are assumed to landfilled. → the landfill process defined in Ecoinvent database, no direct emissions are taken into account. Therefore the existing processes are modified using the “MSW deposition, landfill incl. landfill gas utilisation and leachate treatment, FR,GB,IE,FI,NO mix EU-27” from the ELCD 3 database. The production of unrecycled waste during EoL treatment are assumed to be incinerated. The collected products are assumed the be shredded followed by magnetic and eddy current separation. The metal fractions are further refined to produced recycled material that can serve for new feedstock production.	Waste concrete {RoW} treatment of, inert material landfill Alloc Def, U Scrap steel {RoW} treatment of, inert material landfill Alloc Def, U Inert waste, for final disposal {RoW} treatment of inert waste, inert material landfill Alloc Def, U Waste plastic, consumer electronics {RoW} treatment of, municipal incineration Alloc Def, U Waste glass {RoW} treatment of waste glass, municipal incineration Alloc Def, U Waste glass {Europe without Switzerland} treatment of waste glass, municipal incineration Alloc Def, U Hazardous waste, for incineration {RoW} treatment of hazardous waste, hazardous waste incineration Alloc Def, U Waste electric and electronic equipment {GLO} treatment of, shredding Alloc Def, U Electronics scrap {RoW} treatment of, metals recovery in copper smelter Alloc Def, U